USER MANUAL

Chandra's Atmospheric Composition Explorer-2 (CHACE-2) onboard Chandrayaan-2

Prepared by	Dr. Smitha V. Thampi, Co-PI, CHACE-2 Dr. Dhanya M. B., PM, CHACE-2
Reviewed by	Dr. Tarun Kumar Pant, Head, ITMP, SPL
Approved by	Dr. Radhika Ramachandran, Director, SPL

DOCUMENT CONTROL SHEET

01	**Security Classification:** Unclassified	02	**Distribution:**Free
03	**Document Type:** User Manual	04	**Document:** USER MANUAL for Chandra's Atmospheric Composition Explorer-2 (CHACE-2) onboard Chandrayaan-2
05	**Month and Year of Publication**: November 2019	06	**Collation:** 1.Pages: (including cover page) - 15 2. Figures: 3 3. Tables: 2
07	**Title**: USER MANUAL forChandra's Atmospheric Composition Explorer-2 (CHACE-2) onboard Chandrayaan-2		
08	**Prepared by**: Dr. Smitha V. Thampi, Co-PI, CHACE-2 Dr. Dhanya M. B., PM, CHACE-2		
09	**Reviewed by**: Dr. Tarun Kumar Pant, Head, ITMP/SPL		
10	**Abstract:** This document provides the information about the data, calibration factors, analysis procedures and the caveats about Chandra's Atmospheric Composition Explorer-2 (CHACE-2) payload data.		
11	**Keywords**: Neutral density, Composition, Mass spectrometer, Lunar Exosphere		
12	**Supplementary element**: NIL		
13	**Version**: 1.0, **Date** :**09/10/2019**		
14	**Revision**: 3 Date 12/11/2020		

CHANGE RECORD

Version		**Date**	**Changed paragraph**	**Remarks**
Issue	**Revision**			
1	0	**09/10/2019**		New Document
1	1	**12/11/2019**	Included more detailed descriptions	Revision -1
1	2	**5/10/2020**	Included the corrections suggested by the Science Peer Review Committee	Revision -2
1	3	**12/11/2020**	Included the minor corrections suggested by the Science Peer Review Committee	Revision -3

CONTENTS

Section	Title	Page
1	Introduction	5
2	Application	5
3	Background	5
4	Scientific Objective of CHACE-2	7
5	Chandra's Atmospheric Composition Explorer-2	7
6	Different Modes of Operation	9
7	Derivation of Partial Pressures from CHACE-2 data	10
8	Background Estimation	12
9	Important Caveats	13

1. INTRODUCTION

This document provides the information about the data, calibration factors, analysis procedures and the caveats about Chandra's Atmospheric Composition Explorer-2 (CHACE-2) payload data.

2. APPLICATION

This document serves as a reference to the users of the CHACE-2 data.

3. BACKGROUND

CHandra's Atmospheric Composition Explorer-2 (CHACE-2), is a quadrupole mass spectrometer on board Chandrayaan 2 orbiter. The scientific objective of this instrument is to measure the neutral composition i.e. relative abundances of species of the tenuous lunar exosphere.

The motivation behind the experiment aboard the Chandrayaan-2 mission is to study the pristine lunar exosphere. Moon has a Surface Boundary Exosphere (SBE) which indicates that the local exosphere of the Moon contains direct information about the surface composition and any possible release of gases from the Lunar interior. While the Apollo series of missions by NASA (1969-1972) could measure the total surface pressure and a few atmospheric constituents (noble gases) on the nightside of the moon, there had been no measurement of the dayside lunar atmosphere (due to the saturation of the Lunar Atmospheric Composition Experiment (LACE) mass spectrometer carried in the Apollo-17 mission during lunar days). On 14th November, 2008 the CHandra's Altitudinal Composition Explorer (CHACE) instrument conducted the first-ever measurements of the sunlit lunar neutral atmosphere in the mass range of 1-100 amu during the descent of the Moon Impact Probe (MIP) from orbital height of 100 km with an unprecedented altitudinal and latitudinal coverage and resolution. CHACE has provided the first altitudinal-latitudinal (from ~100 km to the surface) distribution of lunar tenuous atmospheric composition on the dayside. However, the MIP mission had been a one-shot mission, i.e., CHACE provided the data for about 44 minutes only, 20 min before the separation, and 24 min from the time of separation from Chandrayaan-1 orbiter to the time of impact at the lunar surface. Nevertheless, this crucial 44-min data were extremely

useful in furthering our knowledge about the composition of the fragile lunar exosphere. Some of the important findings from the CHACE could be listed as follows:

- First observation of the dayside lunar atmosphere [Sridharan et al.2010b].
- First direct evidence of water vapour in lunar atmosphere [Sridharan et al.2010a].
- Dominance of 44 amu in the mass spectra [Sridharan et al.2010b].
- Significant quantities of heavier species ($\geq$ 60 amu) [Sridharan et al.2013b].
- Signature of spatial heterogeneity and North-South asymmetry of radiogenic activities [Sridharan et al.2013a].
- 2D maps of Argon and H_2 [Thampi et al., 2015]
- 2D maps of Neon [Das et al., 2016]
- Upper limits of lunar Helium during extreme conditions [Das et al., 2017].

Apart from this, in the year 2010 the Lunar Crater Observation and Sensing Satellite (LCROSS) impactor of the Lunar Reconnaissance Orbiter (LRO) mission has analysed the ejecta of the impact and detected, water, other volatile compounds including light hydrocarbons, sulphur-bearing species, and carbon dioxide [Colaprete et al.2010].

The Lunar Atmospheric and Dust Environment Explorer (LADEE) [Elphic et al., 2014] conducted in-situ studies of the equatorial (limited to$\pm 23^{\circ}$ latitude) lunar exosphere from December 2013 to April 2014 covering an altitude range of 61 km down to 4 km. LADEE carried the Neutral Mass Spectrometer (NMS) instrument that studied the lunar exosphere from its equatorial orbit.

Therefore, the next logical step is to envisage a detailed and long term investigation with a CHACE-type mass spectrometer on the lunar polar orbiter: enabling a continuous quantitative measurement of lunar exosphere, especially over the high-latitude/polar region. Due to its large sensitivity and inherent dynamic range which ensured that the instrument remained active in the dynamic range even during the sunlit conditions, CHACE-type instrument in an orbiter would bring out the distinct differences between the day and night sides of the lunar atmosphere. This calls for an orbiter-borne experiment drawing heritage from the CHACE-1.

The unanswered scientific questions, which will be addressed by the CHACE-2 experiment include the long and short term variations of the exospheric density and composition,

especially the diurnal variation and changes during the magnetosphere passage of the moon during each lunation.

4. Scientific Objective of CHACE-2

To characterize lunar neutral exosphere in terms of its composition and density and to understand its variability.

5. Chandra's Atmospheric Composition Explorer-2

The CHACE-2 instrument consists of the sensor probe (cylindrical in shape) and the electronics. The sensor probe consists of a built-in electron impact ionizer along with a Bayard-Alpert collector to measure the total pressure; a set of four quadrupole rods and a detector assembly. The detector assembly consists of a Faraday Cup (FC) and a Channel Electron Multiplier (CEM). In the ionizer section, the ambient neutrals are ionized by energetic electrons. The emission circuit of the ionizer consists of thoriated iridium dual filament system, which emit thermionic electrons when operated in vacuum. The emitted thermionic electrons are post-accelerated to a given energy (typically 70 eV) to ionize the neutrals. A part of the ambient neutrals gets positively ionized within a region in the ionizer called the source grid region. The ion optics potentials drag the ions formed in the ionizer through an orifice to the entry port of the quadrupole mass filter system. Another part of the ambient neutrals gets ionized outside the source grid; they are collected by the Bayard-Alpert gauge collector to constitute a line current, which is calibrated to read the total pressure.

The QMA consists of a set of four conducting parallel cylindrical rods, arranged in equal angular separation of 90°. The pair of rods, which are diametrically opposite, are electrically shorted. Each pair of rods is applied with an RF (frequency 2.3 MHz) potential with a superimposed DC potential. The positive ions, formed within the source grid, are attracted towards the entry aperture of the quadrupole rod system. While at the entry aperture all kinds of positively ionized species are present, only the species with a particular charge-to-mass-ratio will have stable trajectory and reach the detector. The other species, with diff erent charge-to-mass-ratios have unstable trajectories and do not reach the detector. Thus

the QMA functions like a mass filter. In CHACE-2, the pass band of the mass filter is sufficient to achieve unit mass resolution.

The detector assembly of CHACE-2 consists of a Faraday cup and a Channel Electron Multiplier (CEM) detector. By default, the path-stable ions impinge on the Faraday Cup detector to give rise to a detector current. While the Faraday cup detector has no internal gain the CEM detector has got a large gain and it can be biased with different voltages. Figure 1 shows the flight model (FM) of the CHACE-2 instrument, and figure 2 shows its mounting on the Chandrayaan-2 orbiter. Table-1 summarizes its important specifications.



Fig. 1 : FM of the CHACE-2 instrument

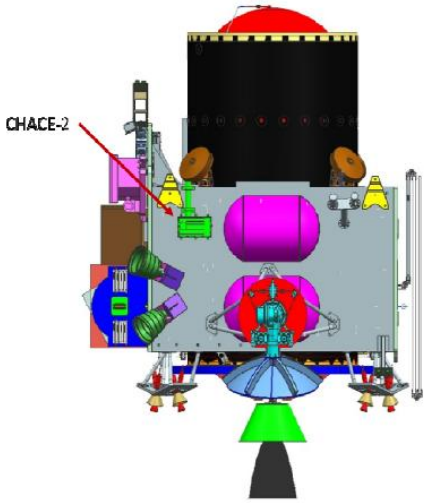


Fig. 2 : Mounting of the CHACE-2 instrument in the Ch-2 orbiter

Table -1 :*Specifications of CHACE-2*

Parameter	**Value**
Mass range	1-300 amu (Programmable)
Mass Resolution	1 amu
Mass filter type	Quadrupole
Detector	Faraday Cup and Channel Electron Multiplier (for composition) along with Beyard Alpert Gauge (for total pressure)
Ionization energy	70 eV (default), programmable.
Dynamic Range	10^{-4} to 10^{-14}Torr
Payload mass	3.73 kg (inclusive of the mass of the DC-DC converter mass of 0.51 kg)
Power budget	33 W raw in nominal mode (69% efficiency of the DC-DC converter under nominal load)
Overall dimensions	235 mm (L) × 108.7 mm (B) × 372 mm

6. Different Modes of Operation

CHACE-2 is envisaged to operate two modes, *viz.* the Mass Sweep mode and the Trend Mode.

Mass Sweep Mode

The mass sweep mode of operation is just described in the previous section where the selected mass range (as per program) is scanned. The data product of this mode of operation is the 'mass spectrum'). This mode is useful to analyze the relative composition of the constituents of a given medium at a given instant ('instant' limits to the 'scan time') of time. This mode is the most commonly exercised mode of CHACE-2. Figure 3 shows a typical mass spectrum obtained during the mass calibration with residual gases in vacuum chamber.

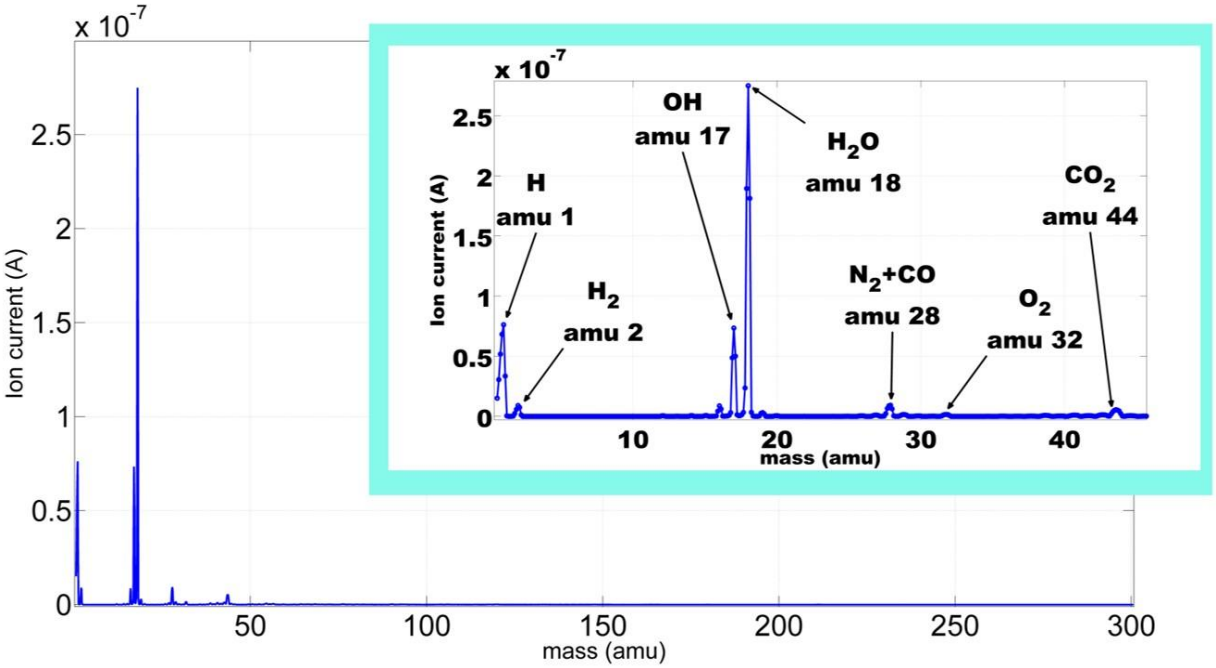


Figure 3: Typical mass spectrum obtained with residual gases in vacuum chamber.

Trend Mode

In the Trend mode, the time-evolution of the relative abundances of only few (limited to the maximum of 15) pre-selected species are tracked. The data product is a time series. The advantage of this mode is good temporal resolution of the data. The time resolution of the relative abundance of a given species in the mass sweep mode is limited by the scan time, which is typically in the order of 10s of seconds, but in the trend mode the time resolution may be as low (lower the better) as milliseconds.

7. Derivation of Partial Pressures from CHACE-2 data

The science data from CHACE-2 are the raw partial pressures (i.e. ion currents for different species with different mass-to-charge ratio, constituting a mass spectrum) and the total pressure. As the first step, the mass spectrum is subjected to two corrections, viz. 1) The Quadrupole Mass Discrimination (QMD) correction and Electron Multiplier Gain (EMG) correction. The effect of QMD and EMG are to compensate the discriminative behaviour of

the QMA and the CEM detector respectively to the species with different mass-to-charge ratios. Until these corrections are carried out, the abundances of the species are incorrect.

Once the QMD and EMG factors are applied to a raw mass spectrum, it presents correct relative abundances.

The corrected mass spectrum is now used to find out the dominant species. In turbo molecular pump based vacuum chambers, under residual gas environment, either water vapour or molecular Nitrogen are the dominant gases. This could be different for any other special environment. Once the dominant gas in the environment is identified, the total pressure reading by the B/A gauge, which is calibrated to read Nitrogen-equivalent pressure, is divided by the relative sensitivity factor of the dominant species. The relative (with respective to molecular Nitrogen) sensitivity factors of the gases are, in general, experimentally measured quantities with the help of absolute pressure sensors like McLeod gauge, which is independent of the type of the dominant gas. Theoretically, it is the ratio of the electron ionization cross sections of the gas of interest to that of molecular Nitrogen.

Once the Nitrogen equivalent total pressure is corrected for the dominant gas, it is used for the normalization. The normalization constant (N) is calculated by dividing the corrected total pressure by the sum over the partial pressures for all the species detected in the mass range of operation.

When the CHACE-2 data are received at the Payload Operation Centre (POC) at SPL, these corrections are applied step-by step using an automated algorithm and the resultant products are archived and distributed as Leve-1 data for further scientific analysis.

In addition to the above steps, the correction for the sensor volume enhancement (Sridharan et al., 2015) and roll – angle correction need to be carried out. The roll angle data are provided in the operating parameter file.

The roll angle correction is in the form $(V_s \cos(\alpha) + v_{th}) / v_{th}$ where V_S is the spacecraft velocity and V_{th} is the thermal velocity of the species. α is the roll angle (Morse et al., 2012) The spacecraft velocity can be derived from SPICE kernels. Table-2 describes the details of the correction factors for the data.

Table-2: Correction factors for the data

Correction factor	**Whether incorporated in the data**	**Remarks**
EMG	Incorporated in the level-1 data before release.	Not to be repeated by user
QMD	Incorporated in the level-1 data before release	Not to be repeated by user
Normalization	Incorporated in the level-1 data before release	Not to be repeated by user
Roll angle correction	To be incorporated by user	Equation given above
Volume enhancement	To be incorporated by user	Division by a factor ~10^4 as described by Sridharan et al (2015)

CHACE-2 Level 1 data

The Level 1 Data Products are the Comma Separated Values (CSV) and xml label files for mass weep mode as well as Trend mode operations. In addition, CSV files and xml label files for HK, operating parameters, and partial pressure data are also given. These CSV files (amu and corrected partial pressures/Total pressure) can be straightaway plotted using any visualization software. To derive the necessary context information, the SPICE toolkit can be used, for which the necessary kernel files are provided in the PDS4 data archive.

8. Background Estimation

The background consists of species that are emanated due to degassing from the instrument as well as spacecraft modules. The mechanisms are thermal desorption, diffusion and

decomposition. This background needs to be carefully estimated from the observations and eliminated to obtain information on the actual lunar constituents. This correction is not done in the Level-1 data product. Users may be able to generate the background model using data from many orbits under different illumination conditions.

It should be noted that while deriving the absolute densities, the dissociation corrections also should be applied. For example, the signal at amu 20 is constituted both by the Neon-20 and doubly ionized Ar-40. Therefore, the user should also look into the Argon signal and estimate the fraction of Ar^{++} that can be present, and subtract these from the observed amu-20 signal to arrive at the partial pressures of Neon-20.

The background corrected partial pressures/ neutral densities would constitute the higher level data products and may be released in due course.

Important Caveats

- Mass spectrometer is a highly sensitive instrument. It can detect the particles of spacecraft outgassing origin, which can never be completely prevented. Although this does not affect the instrument performance, it does affect the limit of detectability for atmospheric neutral particles. Hence, the background needs to be carefully determined and subtracted.
- The mass spectra obtained when the CHANDRAYAAN-2 spacecraft was outside the lunar exosphere can be used to get an estimate of the initial spacecraft background. However, it must be remembered that the background also keeps changing according to the instantaneous instrument orientation, solar illumination, and is a function of time as well as its history of operations. Hence, choice of a proper background requires a very careful understanding of the instrument.
- Moreover, the mass peaks at amu= 18, and 17 could be dominantly contributed by the outgassed water vapour and should not be confused as entirely of Lunar origin. Similarly, the mass peak at 1 amu also contains a significant fraction from the decomposed water. Apart from this any permanent mass peak observed irrespective of latitude, altitude, local time, or any mass peak whose amplitude does not show any

variability with respect to any of the context variables, may be treated as of non-Lunar origin, and should not be interpreted as a Lunar atmospheric constituent.

- Atomic Oxygen, being highly reactive, the observed signal should be properly scaled to get the actual number density. The partial pressures from the data DO NOT correspond to the actual density of atomic O in comparison with any other species. However, this can be used to understand the relative variation of atomic O (day-to-day/seasonal etc.).
- In the interest of a free and open exchange of scientific idea and providing the accurate interpretation to the mass spectrometer data, a few documents will be provided in the PDS archive. Users are advised to go through them carefully before analysing the data to ensure that the inherent limitations of the data are well-understood before interpreting the observations.
- The users are encouraged to contact and discuss with the CHACE-2 team for any clarification.

Contacts:

Dhanya M. B, SPL

mb_dhanya@vssc.gov.in Phone : 91-471-2562553

Smitha V. Thampi, SPL

smitha_vt@vssc.gov.in Phone : 91-471-2562119

References:

- Colaprete, Anthony, Peter Schultz, Jennifer Heldmann, Diane Wooden, Mark Shirley, Kimberly Ennico, Brendan Hermalyn, William Marshall, Antonio Ricco, Richard C, Elphic, et al., Detection of water in the lcross ejecta plume, science, 330(6003), 463–468, 2010
- Elphic, R. C., G. T. Delory, B. P. Hine, P. R. Mahaffy, M. Horanyi, A. Colaprete, M. Benna and S. K. Noble, The Lunar Atmosphere and Dust Environment Explorer Mission, Space Science Review, 185, 3–25, 2014.
- R. Sridharan, S.M.Ahmed,TirthaPratimDas,P.Sreelatha,P. Pradeep Kumar, Neha Naik,Gokulapati Supriya, Corrigendum to "The Sunlit lunar atmosphere: AComprehensive study by CHACE on the Moon Impact Probe of Chandrayaan-1"[Planet. SpaceSci.58(2010)1567–1577], Planetary and Space Science,111167–168, 2015
- Sridharan, R, T.P. Das, SM Ahmed, and A Bhardwaj, Indicators for localized regions of heavier species in the lunar surface from CHACE on Chandrayaan-1, Current Science, 105(11), 1470–1472, 2013b.
- Sridharan,R.,Ahmed,S.M,Das,TirthPratim,Sreelatha,P.,PradeepKumar,P.,Naik,Neha,Supriya ,Gogulapati. 'Direct' evidence for water (H2O) in the sunlit ambience from CHACE on MIP of Chandrayaan I, Planet.SpaceSci.581567–1577,2010a.
- Sridharan, R., S. M. Ahmed, T.P. Das, P. Sreelatha, P. Pradeepkumar, N. Naik, and G. Supriya, The sunlit lunar atmosphere: A comprehensive study by CHACE on the Moon Impact Probe of Chandrayaan-1, Planetary and Space Science, 58, 1567–1577, 2010b.
- Sridharan, R., T. P. Das, S. M. Ahmed, G. Supriya, A. Bhardwaj, and J. A. Kamalakar, Spatial heterogeneity in the radiogenic activity of the lunar interior: Inferences from CHACE and LLRI on Chandrayaan-1, Advances in Space Research, 51, 168–178, 2013a.
- Thampi, S.V., R. Sridharan, T.P. Das, S.M. Ahmed, J.A. Kamalakar, and A. Bhardwaj, The spatial distribution of molecular Hydrogen in the lunar atmosphere – New results, Planetary and Space Science, 106, 142–147, 2015.
- Tirtha Pratim Das, Smitha V. Thampi, M B Dhanya, Anil Bhardwaj, S M Ahmed and R Sridharan, Upper limit of helium-4 in the sunlit lunar exosphere during magnetotail passage under low solar wind condition: Result from CHACE aboard MIP in Chandrayaan-1, Icarus, 297, 189 – 194, 2017.
- Tirtha Pratim Das., S. V. Thampi, A. Bhardwaj, S. M. Ahmed, and R. Sridharan, Observation of Ne at mid and high latitudes in the sunlit Lunar Exosphere: Results from CHACE aboard MIP/Chandrayaan-1, Icarus, 272, 206 – 211, 2016.